

Solutions

Numbering follows the lecture notes. For TYPE C exercises the idea is boxed; the rest are routine once the idea is in hand, which is what the types mean.

Lectures 9–11

§1 Gaussian processes

1.10 A (a) $(Mu)_i = \sum_k M_{ik}u_k$, so $\mathbb{E}[(Mu)_i(Mv)_j] = \sum_{k,l} u_k v_l \mathbb{E}[M_{ik}M_{jl}] = \sum_{k,l} u_k v_l \delta_{ij} \delta_{kl} = \delta_{ij} \langle u, v \rangle$. The pair (Mu, Mv) is a linear image of the Gaussian vector of entries of M , hence jointly Gaussian (Theorem 1.5 of Lectures 1–2), and $\mathbb{E}\langle Mu, Mv \rangle = \sum_i \mathbb{E}[(Mu)_i(Mv)_i] = N \langle u, v \rangle$.

(b) Cauchy–Schwarz: $\langle g, u \rangle \leq \|g\| \|u\| = \|g\|$ for unit u , with equality at $u = g/\|g\|$. $\mathbb{E}\|g\|^2 = \sum_i \mathbb{E}g_i^2 = d$, and Jensen gives $\mathbb{E}\|g\| \leq \sqrt{\mathbb{E}\|g\|^2} = \sqrt{d}$.

(c) $\mathbb{E}\|g\|^4 = \mathbb{E}(\sum_i g_i^2)^2 = \sum_i \mathbb{E}g_i^4 + \sum_{i \neq j} \mathbb{E}g_i^2 \mathbb{E}g_j^2 = 3d + d(d-1) = d^2 + 2d$. Hölder with exponents 3 and $\frac{3}{2}$ applied to $\|g\|^2 = \|g\|^{2/3} \cdot \|g\|^{4/3}$ gives $\mathbb{E}\|g\|^2 \leq (\mathbb{E}\|g\|^{2/3})^{3/2} (\mathbb{E}\|g\|^4)^{1/3}$, so $\mathbb{E}\|g\| \geq (\mathbb{E}\|g\|^2)^{3/2} / (\mathbb{E}\|g\|^4)^{1/2} = d^{3/2} / \sqrt{d^2 + 2d} = \sqrt{d} / \sqrt{1 + 2/d} \geq \sqrt{d}(1 - 1/d) \geq \sqrt{d} - 1$, using $(1+x)^{-1/2} \geq 1 - x/2$. At $d = 100$ the bounds are 9 and 10; the exact value is $\sqrt{2} \Gamma(\frac{101}{2}) / \Gamma(50) = 9.975$.

1.11 A (a) Sum: $\phi(t) = (\phi_1(t), \phi_2(t)) \in \mathbb{R}^{k_1+k_2}$ gives $\langle \phi(s), \phi(t) \rangle = (\Sigma_1)_{s,t} + (\Sigma_2)_{s,t}$. Product: $\phi(t) = (\phi_1(t)_a \phi_2(t)_b)_{a \leq k_1, b \leq k_2} \in \mathbb{R}^{k_1 k_2}$ gives

$$\langle \phi(s), \phi(t) \rangle = \sum_{a,b} \phi_1(s)_a \phi_1(t)_a \phi_2(s)_b \phi_2(t)_b = (\Sigma_1)_{s,t} (\Sigma_2)_{s,t}.$$

A kernel with a feature map is positive semidefinite (Lemma 1.2 applied to the process it defines, or directly: $\sum_{ij} c_i c_j \Sigma_{t_i, t_j} = \|\sum_i c_i \phi(t_i)\|^2 \geq 0$).

(b) $\langle u, v \rangle$ is a kernel on $\mathbb{R}^d$ with $\phi = \text{id}$; by (a), so is every power $\langle u, v \rangle^m$, with feature map $u \mapsto u^{\otimes m} = (u_{a_1} \cdots u_{a_m})_{a_1, \dots, a_m} \in \mathbb{R}^{d^m}$. (So is every polynomial in $\langle u, v \rangle$ with nonnegative coefficients, and in the limit $e^{\langle u, v \rangle}$ — the kernel behind the Gaussian-kernel machines of statistics.)

(c) The feature map for $\langle u, v \rangle^2$ from (a) is $\phi(u) = (u_a u_b)_{a,b} = uu^\top$, a $d \times d$ matrix, with the Frobenius inner product. The process is $X_u = \langle uu^\top, G \rangle = \sum_{a,b} u_a u_b G_{ab} = u^\top G u$ for G a $d \times d$ matrix with i.i.d. $\mathcal{N}(0, 1)$ entries. Check: $\mathbb{E}[u^\top G u v^\top G v] = \sum_{a,b,c,e} u_a u_b v_c v_e \delta_{ac} \delta_{be} = \langle u, v \rangle^2$.

1.12 A (a) $\mathbf{1}_{[0,s]} \mathbf{1}_{[0,t]} = \mathbf{1}_{[0, \min(s,t)]}$, whose integral is $\min(s, t)$.

(b) (B_r, B_s, B_t) is a Gaussian vector with covariance given by $\min$, so $B_t - B_s$ is Gaussian with variance $\Sigma_{t,t} - 2\Sigma_{s,t} + \Sigma_{s,s} = t - 2s + s = t - s$, and $\text{Cov}(B_t - B_s, B_r) = \min(t, r) - \min(s, r) = r - r = 0$ for $r \leq s$. Jointly Gaussian plus uncorrelated is independent, Theorem 1.16 of Lectures 1–2. (Through the feature map: $B_t - B_s = \langle \mathbf{1}_{(s,t]}, g \rangle$, and $\mathbf{1}_{(s,t]} \perp \mathbf{1}_{[0,r]}$.)

1.13 B (a) Theorem 2.9 of Lectures 1–2 with $\Sigma_{t,t} = t$, $\Sigma_{t,1} = \min(t, 1) = t$, $\Sigma_{1,1} = 1$: $\mathbb{E}[B_t | B_1] = \Sigma_{t,1} \Sigma_{1,1}^{-1} B_1 = t B_1$ and $\text{Var}(B_t | B_1) = t - t^2 = t(1 - t)$. So $B_t = t B_1 + (B_t - t B_1)$ with the bridge $B_t - t B_1$ independent of B_1 , Gaussian with variance $t(1 - t)$: zero at both ends, largest in the middle.

(b) Let $t \in [t_j, t_{j+1}]$ (with $t_0 = 0$, $B_0 = 0$) and let $L_t = B_{t_j} + \frac{t-t_j}{t_{j+1}-t_j} (B_{t_{j+1}} - B_{t_j})$ be the linear interpolation. By Theorem 2.9 of Lectures 1–2 the conditional mean is the unique linear combination $\sum_i a_i B_{t_i}$ with $B_t - \sum_i a_i B_{t_i}$ uncorrelated with every B_{t_k} ; L_t is such a combination, so it suffices to check $\text{Cov}(B_t - L_t, B_{t_k}) = 0$ for each k . Write $\theta = \frac{t-t_j}{t_{j+1}-t_j}$, so $B_t - L_t = (1 - \theta)(B_t - B_{t_j}) - \theta(B_{t_{j+1}} - B_{t_j})$. For $k \leq j$, both increments are independent of B_{t_k} by Exercise 1.12(b). For $k \geq j+1$, $\text{Cov}(B_t - B_{t_j}, B_{t_k}) = t - t_j$ and $\text{Cov}(B_{t_{j+1}} - B_{t_j}, B_{t_k}) = t_{j+1} - t_j$, so the covariance is $(1 - \theta)(t - t_j) - \theta(t_{j+1} - t_j) = 0$ by the choice of θ . (For t beyond t_n the same computation with no right endpoint gives $\mathbb{E}[B_t | \cdots] = B_{t_n}$.) The conditional variance is $\theta(1 - \theta)(t_{j+1} - t_j)$ by (a) rescaled, so the posterior of Brownian motion given n points is a chain of independent bridges — exactly the picture one draws.

1.14 C (a) For each g , $\sup_t X_t = \sup_t \langle \phi(t), g \rangle = \sup_{h \in S} \langle h, g \rangle$: the two suprema run over the same set of numbers. Take expectations.

(b) For fixed $h_0 \in S$, $\sup_{h \in S} \langle h, g \rangle \geq \langle h_0, g \rangle$, whose expectation is 0; monotonicity in S is monotonicity of the supremum. $\langle h + x, g \rangle = \langle h, g \rangle + \langle x, g \rangle$ with $\mathbb{E}\langle x, g \rangle = 0$, and $\langle \lambda h, g \rangle = \lambda \langle h, g \rangle$. A linear function attains its supremum over the convex hull at a point of the set: if $h = \sum_j \theta_j h_j$ with $\theta_j \geq 0$, $\sum_j \theta_j = 1$, then $\langle h, g \rangle = \sum_j \theta_j \langle h_j, g \rangle \leq \max_j \langle h_j, g \rangle$, so $w(\text{conv } S) \leq w(S)$, and $\geq$ is monotonicity.

(c) The supremum of $\langle h, g \rangle$ over a unit ball is the dual norm: $\sup_{\|h\|_2 \leq 1} \langle h, g \rangle = \|g\|_2$ (Cauchy–Schwarz, equality at $h = g/\|g\|$), $\sup_{\|h\|_1 \leq 1} \langle h, g \rangle = \max_i |g_i|$ (attained at $h = \pm e_i$), $\sup_{\|h\|_\infty \leq 1} \langle h, g \rangle = \|g\|_1$ (attained at $h = \text{sign}(g)$). Then: $\sqrt{k} - 1 \leq \mathbb{E}\|g\|_2 \leq \sqrt{k}$ is Exercise 1.10(c). $\max_i |g_i| = \max_{j \leq 2k} Y_j$ with $Y = (g_1, \dots, g_k, -g_1, \dots, -g_k)$, $2k$ standard Gaussians, dependent, and Proposition 1.9 does not care: $\leq \sqrt{2 \log(2k)}$. $\mathbb{E}\|g\|_1 = k \mathbb{E}|g_1| = k\sqrt{2/\pi}$, since $\mathbb{E}|g_1| = 2 \int_0^\infty x e^{-x^2/2} dx / \sqrt{2\pi} = \sqrt{2/\pi}$.

(d) For $h \in S_k$ with support I , $|I| \leq k$, Cauchy–Schwarz on the coordinates in I gives $\langle h, g \rangle \leq \|h\|_2 (\sum_{i \in I} g_i^2)^{1/2} \leq (\sum_{i \in I} g_i^2)^{1/2}$, with equality when h is proportional to $(g_i)_{i \in I}$; maximizing over I picks the k largest $|g_i|$, so $\sup_{h \in S_k} \langle h, g \rangle = (\sum_{i \leq k} |g|_{(i)}^2)^{1/2}$ with $|g|_{(1)} \geq |g|_{(2)} \geq \dots$ the order statistics of $|g_i|$. This is at most $\sqrt{k} \max_i |g_i|$, and (c) gives $w(S_k) \leq \sqrt{k} \sqrt{2 \log(2d)}$. Against $w(B_d^2) \geq \sqrt{d} - 1$: the sparse set has width of order $\sqrt{k \log d}$, as if it lived in $k \log d$ dimensions. The number of Gaussian measurements needed to recover a k -sparse vector is, up to constants, the square of its width — $k \log d$ rather than d — which is the content of the sharper bound $w(S_k) \leq \sqrt{2k \log(ed/k)}$ and of Gordon’s escape theorem.

1.15 B (a) $\text{Cov}(X_i, X_j) = \rho + (1 - \rho)\delta_{ij}$: all variances 1, all correlations ρ . Since g_0 does not depend on i , $\max_i X_i = \sqrt{\rho} g_0 + \sqrt{1 - \rho} \max_i g_i$, and $\mathbb{E}g_0 = 0$ gives $\mathbb{E} \max_i X_i = \sqrt{1 - \rho} \mathbb{E} \max_i g_i \approx \sqrt{(1 - \rho)2 \log n}$. The union bound $\sqrt{2 \log n}$ is attained only at $\rho = 0$; at $\rho = 1$ all X_i coincide and the maximum is a single $\mathcal{N}(0, 1)$ with mean 0.

(b) The union bound (Proposition 1.9, $\text{Var } X_i = \|u_i\|^2 = 1$) gives $\sqrt{2 \log n}$. For the other bound, $\max_i \langle u_i, g \rangle \leq \sup_{\|u\|=1} \langle u, g \rangle = \|g\|$ pointwise, so $\mathbb{E} \max_i X_i \leq \mathbb{E}\|g\| \leq \sqrt{d}$ (Exercise 1.10). The first bound is sharp for $n \leq d$ orthonormal u_i : then the X_i are i.i.d. standard and Proposition 2.3 gives $\sqrt{2 \log n}(1 + o(1))$. The second is sharp when the u_i are spread over the whole sphere: if every unit vector is within ε of some u_i , then the maximizer $u^* = g/\|g\|$ has a u_i with $\|u_i - u^*\| \leq \varepsilon$, so $\max_i X_i \geq \langle u^*, g \rangle - \varepsilon \|g\| = (1 - \varepsilon)\|g\|$ and $\mathbb{E} \max_i X_i \geq (1 - \varepsilon)\mathbb{E}\|g\|$. Lecture 10 counts how many points that takes: $e^{\Theta(d)}$, which is where the two bounds cross.

1.16 B (a) $X_t = \langle \phi(t), g \rangle = 0$ iff $\phi(t) \perp g$ iff $\phi(t) \in g^\perp$; since $g \neq 0$ a.s., $g^\perp$ is a hyperplane through the origin. For a rotation U , $Ug \sim \mathcal{N}(0, UU^\top) = \mathcal{N}(0, I_k)$ (Theorem 1.5 of Lectures 1–2), and $(Ug)^\perp = U(g^\perp)$, so $U(g^\perp)$ has the same law as $g^\perp$. Equivalently: the unit normal $g/\|g\|$ is uniform on S^{k-1} .

(b) By (a), $\#\{t : X_t = 0\}$ is the number of times the curve ϕ — equivalently its projection γ , since $\phi(t) \in g^\perp$ iff $\gamma(t) \in g^\perp$ — crosses the random hyperplane $g^\perp$. Let $f(A)$ be the expected number of crossings for a curve A on the sphere. It is additive when a curve is cut into pieces (a random hyperplane misses any given finite set of points with probability one), and by (a) it is invariant under rotations. All arcs of great circles of length ℓ are rotations of one another, so f of such an arc is a function $F(\ell)$ of the length alone; it is additive, $F(\ell + \ell') = F(\ell) + F(\ell')$, and nondecreasing, hence $F(\ell) = c\ell$. A full great circle C , of length 2π , is the unit circle of a 2-plane P ; the hyperplane $g^\perp$ meets P in a line through the origin (with probability one it does not contain P), and a line through the center meets a circle in exactly two points. So $2 = F(2\pi) = 2\pi c$ and $c = 1/\pi$. For a smooth γ , inscribe great-circle polygons: their lengths tend to $\text{length}(\gamma)$ and their crossing counts to those of γ , giving $\mathbb{E}\# = \text{length}(\gamma)/\pi$. This is Crofton’s formula; in the plane, the same argument is Buffon’s needle.

(c) $X_\theta = \langle \phi(\theta), g \rangle$ with $\phi(\theta) = (\cos \theta, \sin \theta, \cos 2\theta, \sin 2\theta, \dots, \cos n\theta, \sin n\theta) \in \mathbb{R}^{2n}$ and $g = (a_1, b_1, \dots, a_n, b_n)$. Then $\|\phi(\theta)\|^2 = \sum_{j \leq n} (\cos^2 j\theta + \sin^2 j\theta) = n$, constant, so $\gamma = \phi/\sqrt{n}$ and $\|\gamma'(\theta)\|^2 = \frac{1}{n} \sum_{j \leq n} j^2 = \frac{(n+1)(2n+1)}{6}$. By (b),

$$\mathbb{E}\#\{\theta \in [0, 2\pi) : X_\theta = 0\} = \frac{1}{\pi} \cdot 2\pi \sqrt{\frac{(n+1)(2n+1)}{6}} = 2\sqrt{\frac{(n+1)(2n+1)}{6}} \sim \frac{2n}{\sqrt{3}}.$$

The feature curve winds around a sphere of radius $\sqrt{n}$ in $\mathbb{R}^{2n}$ at constant speed, and a random hyperplane cuts it about $2n/\sqrt{3}$ times — a bit more than the 2 cuts of the great circle $n = 1$ per unit of frequency, because higher harmonics make the curve longer. (For comparison, the real zeros of a random *algebraic* polynomial of degree n with i.i.d. Gaussian coefficients number only $\sim \frac{2}{\pi} \log n$: the feature curve $(1, x, \dots, x^n)$, projected to the sphere, is short.)

§2 Suprema

2.14 A (a) Put $t_j = \min\{(2j-1)\varepsilon^2, 1\}$ for $j = 1, \dots, \lceil 1/(2\varepsilon^2) \rceil$. Every $t \in [0, 1]$ is within ε^2 of some t_j in $|\cdot|$, hence within ε in $d = \sqrt{|\cdot|}$, so $N([0, 1], d, \varepsilon) \leq \lceil 1/(2\varepsilon^2) \rceil \leq 1 + 1/(2\varepsilon^2)$. The diameter is 1, and Theorem 2.9 gives $\mathbb{E} \sup_{t \leq 1} B_t \leq 12 \int_0^{1/2} \sqrt{\log(1 + 1/(2\varepsilon^2))} d\varepsilon = 12 \times 0.806 = 9.68$, finite. The exact value, $\sqrt{2/\pi}$, comes from the reflection principle, $\sup_{t \leq 1} B_t \stackrel{d}{=} |B_1|$ — a statement about paths, outside this course.

(b) Points $t_j = \min\{(2j-1)\varepsilon/L, 1\}$, $j \leq \lceil L/(2\varepsilon) \rceil$, are within ε/L of every t , hence within ε in d ; so $N \leq 1 + L/(2\varepsilon)$. Also $d(s, t) \leq \sqrt{\text{Var } X_s} + \sqrt{\text{Var } X_t} \leq 2$, so the integral in Theorem 2.9 runs over at most $[0, 1]$. For $\varepsilon \leq 1$,

$1 + L/(2\varepsilon) \leq (1 + L)/\varepsilon$, so with $A = 1 + L$ the given estimate yields $\mathbb{E} \sup X_t \leq 12(\sqrt{\log(1 + L)} + 1)$. (The estimate itself: $\sqrt{\log(A/\varepsilon)} \leq \sqrt{\log A} + \sqrt{\log(1/\varepsilon)}$ and $\int_0^1 \sqrt{\log(1/\varepsilon)} d\varepsilon = \Gamma(\frac{3}{2}) < 1$.) The order $\sqrt{\log L}$ is that of the maximum of L independent standard Gaussians: a process that changes on scale $1/L$ has about L genuinely different values on $[0, 1]$. Check: the kernel $\Sigma_{s,t} = e^{-L^2(s-t)^2/2}$ has $d(s,t)^2 = 2(1 - e^{-L^2(s-t)^2/2}) \leq L^2(s-t)^2$; at $L = 50$ a simulation gives $\mathbb{E} \sup = 2.27$, against $\sqrt{2 \log L} = 2.80$ and the bound 34.

(c) That nearby points carry nearly the same value: the union bound treats $B_{j/n}$ and $B_{(j+1)/n}$ as two unrelated Gaussians, while chaining bounds their *difference*, whose standard deviation $1/\sqrt{n}$ is small, and pays $\sqrt{\log(\text{number of links})}$ only against that small scale.

2.15 B (a) Let $1 - \varepsilon \leq \|x\| \leq 1$, so $x \neq 0$, and put $u = x/\|x\| \in S^{d-1}$. Some u_i has $\|u - u_i\| \leq \varepsilon$, and $\|x - u\| = 1 - \|x\| \leq \varepsilon$, so $\|x - u_i\| \leq 2\varepsilon$. Hence the n balls $B(u_i, 2\varepsilon)$ cover the shell $\{1 - \varepsilon \leq \|x\| \leq 1\}$, whose volume is $(1 - (1 - \varepsilon)^d) \text{vol}(B_2^d)$; the balls have total volume $n(2\varepsilon)^d \text{vol}(B_2^d)$. So $n(2\varepsilon)^d \geq 1 - (1 - \varepsilon)^d$.

(b) At $\varepsilon = \frac{1}{4}$: $n \geq 2^d(1 - (3/4)^d) \geq 2^d(1 - \frac{9}{16}) = \frac{7}{16}2^d \geq 2^{d-2}$ for $d \geq 2$. With Lemma 2.6, $2^{d-2} \leq N(S^{d-1}, \|\cdot\|, \frac{1}{4}) \leq 9^d$: the covering number of the sphere at a fixed scale is $e^{\Theta(d)}$.

(c) Sudakov's inequality at $\varepsilon = \frac{1}{4}$ with (b): $\mathbb{E}\|g\| = \mathbb{E} \sup_u \langle g, u \rangle \geq c \cdot \frac{1}{4} \sqrt{\log 2^{d-2}} = \frac{c}{4} \sqrt{(d-2) \log 2}$, which is $c' \sqrt{d}$ for $d \geq 4$, say. Example 2.11 gives $\mathbb{E}\|g\| \leq 12c_0 \sqrt{d}$ from the upper bound $N \leq (1 + 2/\varepsilon)^d$. So $\mathbb{E}\|g\| \asymp \sqrt{d}$ follows from $N(S^{d-1}, \|\cdot\|, \varepsilon) = e^{\Theta(d)}$ at a fixed scale, and from nothing else about the sphere. (Of course Exercise 1.10 gives the sharp $[\sqrt{d} - 1, \sqrt{d}]$ directly; the point is that the covering numbers alone already know the order.)

2.16 B (a) $Y_{(u,v)} = \sum_{ij} v_i u_j M_{ij} = \langle vu^\top, M \rangle_F$ is linear in the entries of M , and for a fixed matrix A , $\text{Var} \langle A, M \rangle_F = \sum_{ij} A_{ij}^2 = \|A\|_F^2$; so $d((u,v), (u',v'))^2 = \text{Var}(\langle vu^\top - v'(u')^\top, M \rangle_F) = \|vu^\top - v'(u')^\top\|_F^2$. Expanding, with $\|ab^\top\|_F = \|a\| \|b\|$ and $\langle ab^\top, cd^\top \rangle_F = \langle a, c \rangle \langle b, d \rangle$, this is $1 + 1 - 2\langle v, v' \rangle \langle u, u' \rangle$. For the triangle bound write $vu^\top - v'(u')^\top = v(u - u')^\top + (v - v')(u')^\top$, so $d \leq \|v\| \|u - u'\| + \|v - v'\| \|u'\| = \|u - u'\| + \|v - v'\|$.

(b) Let U be an $\varepsilon/2$ -net of S^{d-1} and V an $\varepsilon/2$ -net of S^{N-1} , of the sizes in Lemma 2.6. For $(u,v) \in T$ pick $u' \in U$, $v' \in V$ within $\varepsilon/2$ of u, v ; by (a), $d((u,v), (u',v')) \leq \varepsilon$. So $U \times V$ is an ε -net, of size at most $(1 + 4/\varepsilon)^d (1 + 4/\varepsilon)^N$. And $D \leq \|vu^\top\|_F + \|v'(u')^\top\|_F = 2$. Theorem 2.9:

$$\mathbb{E}\|M\|_{\text{op}} = \mathbb{E} \sup_T Y \leq 12 \int_0^1 \sqrt{(N+d) \log(1 + 4/\varepsilon)} d\varepsilon = 12c_1 \sqrt{N+d}, \quad c_1 = 1.559,$$

and $12c_1 = 18.7 \leq 19$, $\sqrt{N+d} \leq \sqrt{N} + \sqrt{d}$. For $N = 100$, $d = 50$ the bound is 229 against the truth 16.6.

(c) By (a) and $|\langle v, v' \rangle| \leq 1$, $d^2 \geq 2 - 2|\langle u, u' \rangle| = \min(\|u - u'\|^2, \|u + u'\|^2)$. Fix any $u \in S^{d-1}$ and any v ; some net point (u_i, v_i) is within ε of (u, v) , so u is within ε of u_i or of $-u_i$. Hence the $2n$ points $\pm u_i$ form an ε -net of S^{d-1} , $2n \geq N(S^{d-1}, \varepsilon)$, and by symmetry $2n \geq N(S^{N-1}, \varepsilon)$. At $\varepsilon = \frac{1}{4}$, Exercise 2.15(b) gives $n \geq \frac{1}{2} 2^{\max(N,d)-2} = 2^{\max(N,d)-3}$.

(d) Sudakov's inequality at $\varepsilon = \frac{1}{4}$: $\mathbb{E}\|M\|_{\text{op}} \geq \frac{c}{4} \sqrt{(\max(N,d) - 3) \log 2} \geq c' \sqrt{\max(N,d)} \geq \frac{c'}{2} (\sqrt{N} + \sqrt{d})$ for $\max(N,d) \geq 6$, say (and $\mathbb{E}\|M\|_{\text{op}} \geq \mathbb{E}|M_{11}| > 0$ covers the rest). With (b), $c''(\sqrt{N} + \sqrt{d}) \leq \mathbb{E}\|M\|_{\text{op}} \leq 19(\sqrt{N} + \sqrt{d})$: the order of the operator norm of a Gaussian matrix follows from the covering numbers of $S^{N-1} \times S^{d-1}$ alone. Lecture 11 gives the constant 1 in the upper bound by comparison.